

HACKORBIT 2K26

Placement Skill-Gap Hub

Final Technical Report

TEAM CODEAVENGERS

Track A — DataDrishti • BANNARI AMMAN INSTITUTE OF TECHNOLOGY

**From Placement Statistics
to Skill-Gap Intelligence.**

Sample / Demonstration Dataset — no real institutional student, placement, or recruiter data is used.

1. Executive Summary & Problem

One clear story from placement outcomes to training action

Placement Skill-Gap Hub is a Power BI decision-support solution that connects placement outcomes, industry skill demand, and student skill readiness into one actionable workflow. The core metric is the skill gap, which is classified by priority and translated into a plain-language training recommendation.

Official problem statement: “Placement Skill-Gap Hub — Model placement stats across branches.”

Traditional placement dashboards are useful for reporting placement rate, average package, companies, and branch performance. The gap is that they do not connect those outcomes to recruiter skill requirements or student readiness.

PLACEMENT OUTCOMES → INDUSTRY SKILL DEMAND → STUDENT SKILL READINESS → SKILL GAP → PRIORITY → TRAINING ACTION

Core question

Which skills should this institution prioritize to improve placement readiness?

The current MVP is deliberately scoped to a two-day build: a three-table demonstration dataset, three Power BI pages, core DAX measures, a Skill Gap Priority Matrix, a rule-based Training Recommendation Engine, and a branch × skill heatmap add-on.

2. Solution & Differentiation

Why this is more than a conventional placement dashboard

Conventional dashboard	Placement Skill-Gap Hub
Reports placement outcomes	Connects outcomes, demand, and readiness
Primarily descriptive	Adds diagnostic skill-gap analysis
Stops at charts	Ends with training priority
One major data layer	Three linked analytical layers

Defensible positioning: our differentiation is the integration of placement outcomes, recruiter skill demand, and student readiness into a single actionable skill-gap workflow. We do not claim this has never been built before.

Rule-based by design

The current recommendation engine is explicitly rule-based. It uses transparent thresholds and fixed recommendation templates; it is not AI or machine learning.

3. Dataset & Data Model

Small, explainable, hackathon-feasible architecture

Table	Key fields
Students_Placement	Student_ID, Branch, Batch, Placement_Status, Company, Package
Student_Skills	Student_ID, Skill, Assessment_Score (0–100)
Industry_Skill_Demand	Skill, Demand_Level (0–100), Companies_Requiring_Count

The model uses two active, single-direction relationships: Students_Placement[Student_ID] → Student_Skills[Student_ID], and Industry_Skill_Demand[Skill] → Student_Skills[Skill]. This keeps the model small and easy to debug and explain.

Demonstration scale: 240 students • 5 branches • 6 skills • 22 sample companies

Data transparency: Demand_Level and Assessment_Score use the same 0–100 scale for arithmetic comparison. They are treated as comparable proxies for a readiness gap, not as identical measurements.

4. Analytics Methodology & Core DAX

Transparent calculations that a judge can verify

Measure	Definition
Placement Rate	Students Placed ÷ Total Students
Average Package	Average package among placed students
Company Count	Distinct recruiting companies
Student Readiness %	Average Assessment_Score in the current skill/branch/batch context
Industry Demand %	Demand_Level for the selected skill
Skill Gap %	Industry Demand % – Student Readiness %
Skill Priority	CRITICAL ≥25 pts; HIGH ≥15; MEDIUM ≥5; otherwise LOW

Core formula: Skill Gap = Industry Demand – Student Readiness

Priority cut-points are deliberate, round-number starting thresholds rather than statistically derived boundaries. A real deployment would tune them against validated institutional outcome data.

5. Dashboard & Signature Features

Three pages, each answering one decision question

Dashboard page	Primary question	Core visuals
Placement Command Center	What is happening?	KPIs, branch placement, company hiring, package distribution
Skill Demand vs. Readiness	What does industry need, and are students ready?	Demand ranking, paired demand/readiness, gap table
Action Center	What should we do next?	Priority Matrix, Recommendation Engine, heatmap

Signature Feature 1 — Skill Gap Priority Matrix

Student Readiness is the X-axis and Industry Demand is the Y-axis. High demand + low readiness is the critical training quadrant.

Signature Feature 2 — Training Recommendation Engine

Each skill receives a gap, priority, and short rule-based recommendation. The completed Branch × Skill Heatmap localizes the largest gaps to a branch.

6. UX, Interactivity & Showcase

Modern presentation layer for the same analytical logic

The web showcase keeps the same data and core logic as the Power BI build while making the live demonstration easier to understand and operate.

- Interactive Branch, Batch, and Skill slicer buttons with clear active states.
- Unified filtering so branch/batch selections scope the skill-gap story as well as placement KPIs.
- Highest Priority Skill insight card that recalculates with the current filter context.
- Live Skill Gap formula display using the current top-priority skill.
- Branch Skill-Gap Heatmap for localized training priorities.
- Animated intro, modern micro-interactions, filter feedback, scroll progress, and quick evaluator demo path.
- Accessibility considerations including focus states and reduced-motion support.

15-second test: a first-time viewer should quickly identify placement health, the critical skill gap, the branch needing attention, and the next action.

7. Impact & Future Scope

Who benefits and what comes next

User	Value
Students	A ranked view of skills to build before placement season.
Placement / Training Team	A prioritized, evidence-based training plan.
Institution Leadership	A clearer basis for training budget and faculty time allocation.

Current MVP

- Three-table demonstration dataset
- Rule-based priority classification
- Manually curated skill list
- Static demonstration data
- Generic recommendation text by priority tier

Future scope

- Real institutional placement and recruiter-data integration
- ML-based placement prediction
- Automated skill extraction from resumes and job descriptions
- Continuous industry-demand updates
- Personalized student learning paths and training-impact simulation

Future capabilities are explicitly separated from what is working today.

8. Testing & Final Implementation Status

What was actually built, tested, and fixed

The final status records the dataset, all three Power BI pages, signature features, web dashboard, and testing as complete. Values were cross-checked against the raw dataset and the dashboard was exercised through real filter interactions.

Issue found	Fix
Web dashboard blank under restrictive networks	Removed external chart dependencies; visuals use local SVG rendering.
Placement Rate filter inconsistency	Explicit filter logic added so numerator and denominator use the intended context.
Skill Gap display scaling	Demand and readiness normalized before percentage formatting.
Industry Demand not changing with Skill	Corrected the field to the Industry_Skill_Demand source.
Heatmap layout width	Corrected markup so the heatmap matches the dashboard container.

Final verified build: 240 students • 5 branches • 6 skills • 22 sample companies • 3 Power BI pages • Priority Matrix • Recommendation Engine • Branch Skill-Gap Heatmap • web companion.

9. Judge Defense, Demo Flow & Conclusion

A compact final-defense sheet

Judge question	Answer
Where did the data come from?	Synthetic demonstration data; no real institutional data is claimed.
Is this AI?	No. The MVP uses a rule-based SWITCH / threshold engine.
How is the gap calculated?	Industry Demand – Student Readiness.
Does it prove causation?	No. It identifies actionable gaps; causal validation is future work.
How is it different?	It connects outcomes, demand, readiness, gap, and action in one workflow.
Can it scale?	Yes, by replacing the demonstration source with validated institutional data and integrations.
Recommended live demo	

Placement Command Center → select Branch/Batch → Skill Demand vs. Readiness → show the largest gap → Action Center → show the priority and recommendation.

We don't just show placement numbers. We show what skills need to improve. Placement Skill-Gap Hub — From Placement Statistics to Skill-Gap Intelligence.

GitHub: codeavengers-placement-skill-gap • Vercel: codeavengers-placement-skill-gap.vercel.app

III.01

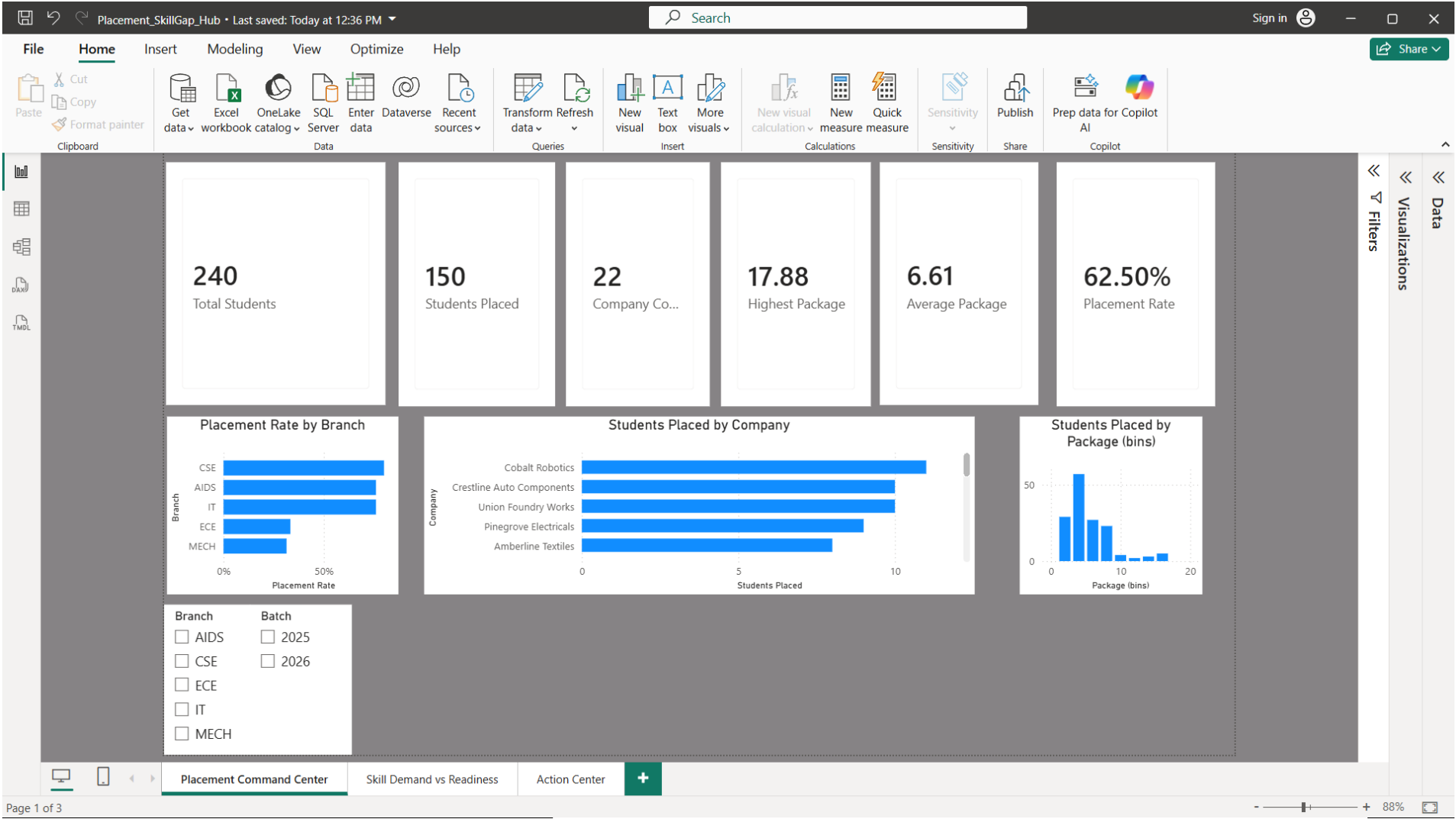
Power BI Desktop

The actual analytics implementation — DAX, data model, native slicers

-
- Placement Command Center
 - Skill Demand vs. Readiness
 - Action Center — Priority Matrix & Recommendation Engine

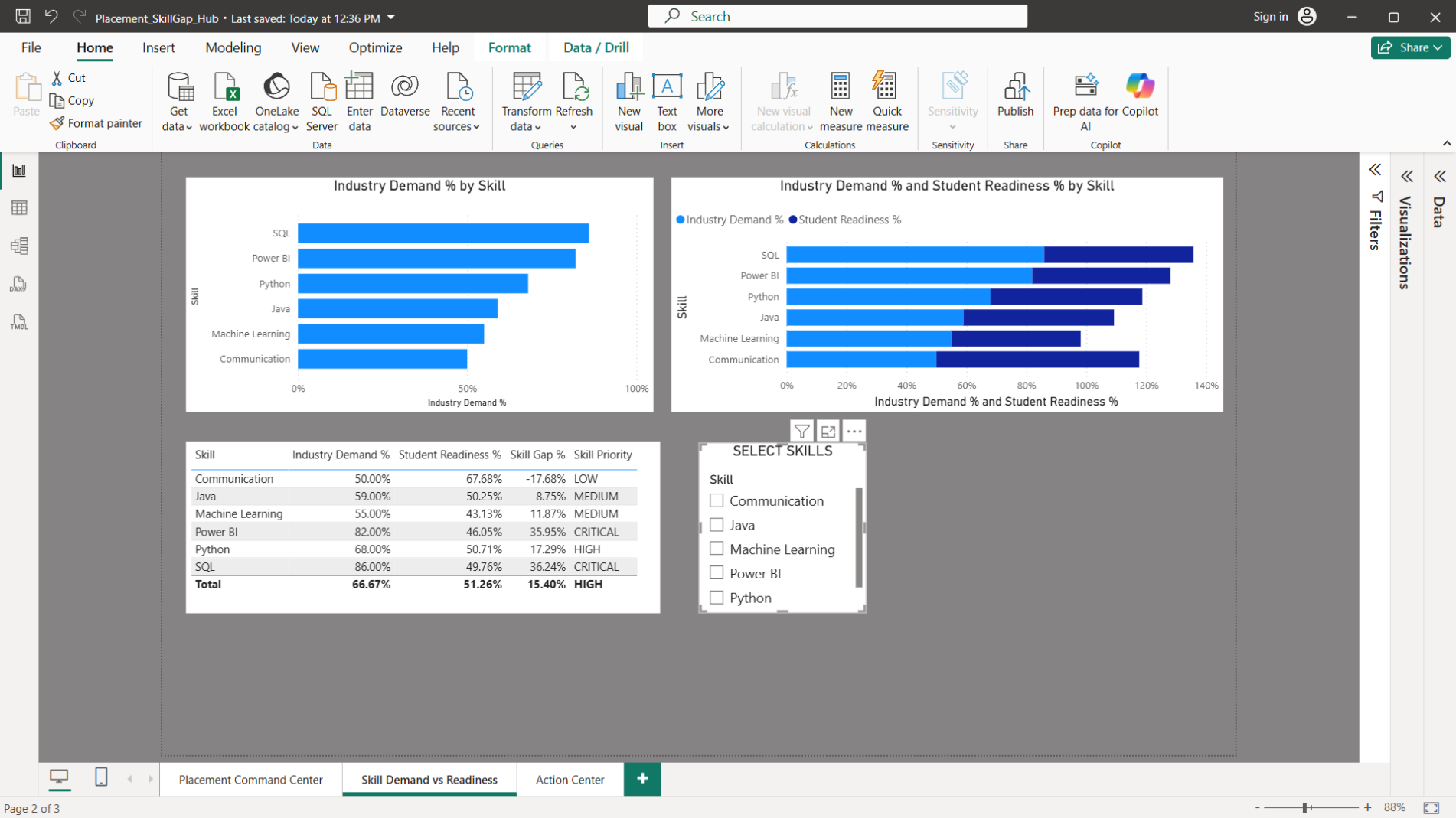
POWER BI DESKTOP

Placement Command Center



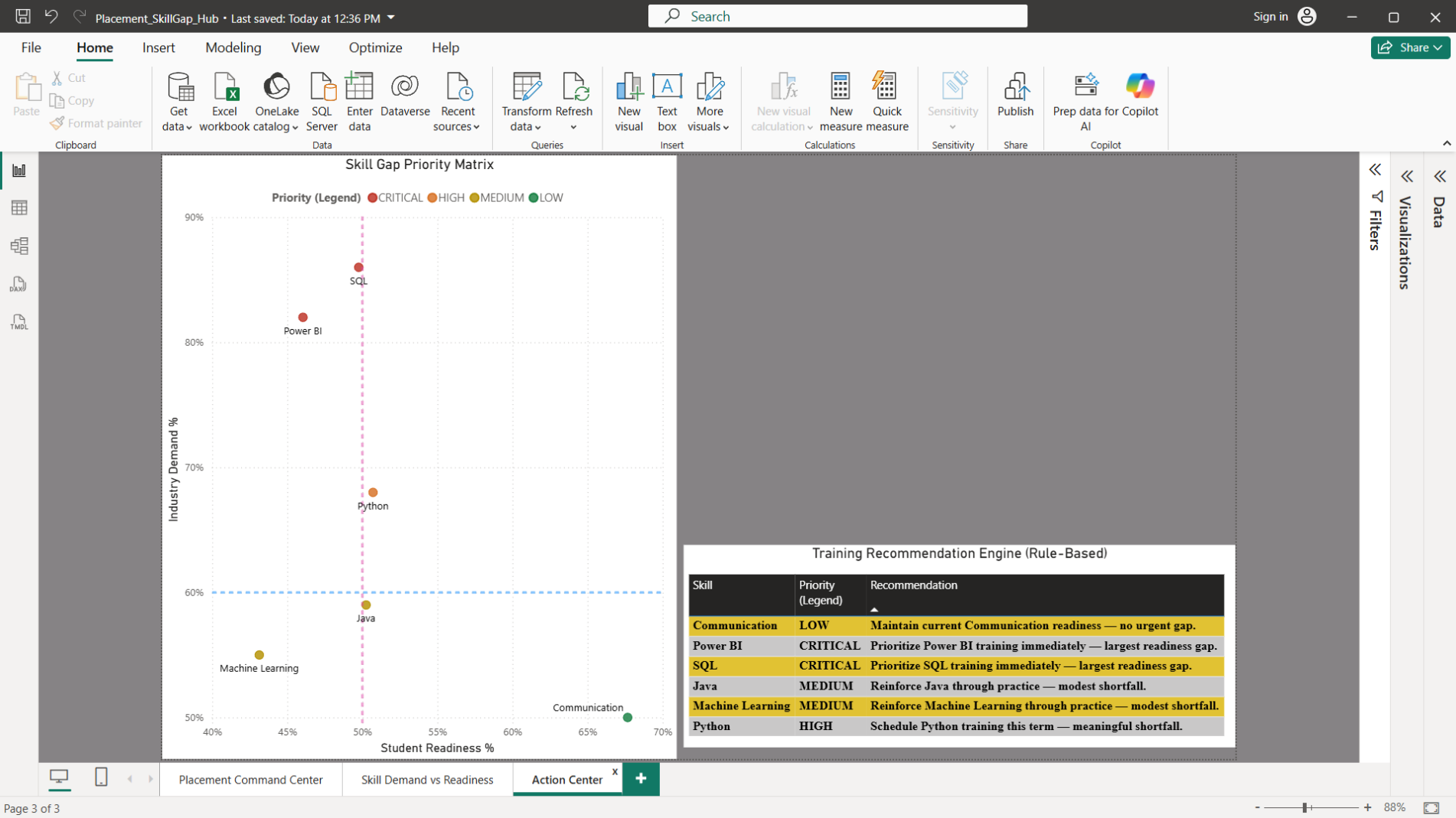
POWER BI DESKTOP

Skill Demand vs. Readiness



POWER BI DESKTOP

Action Center — Priority Matrix & Recommendation Engine



III.02

Web Dashboard

The live companion site — zero dependencies, same data and logic as Power BI

-
- Placement Command Center
 - Skill Demand vs. Readiness
 - Action Center — Priority Matrix & Recommendation Engine

Placement Command Center

Placement Skill-Gap Hub
Codeavengers · HACKORBIT 2K26

Command Center

Skill Gap

Action Center

SAMPLE / DEMONSTRATION DATASET

01 Placement Command Center

What is happening?

BRANCH

AIDS

CSE

ECE

IT

MECH

BATCH

2025

2026

[Reset](#)

Showing all 240 students

● All branches · All batches

● Use slicer buttons to drive the entire decision story.

240

Total Students

150

Students Placed

62.5%

Placement Rate

6.61

Avg Package (LPA)

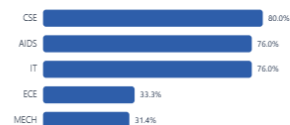
17.88

Highest Package (LPA)

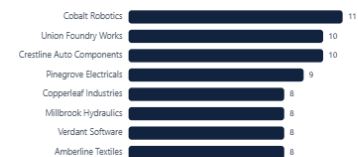
22

Company Count

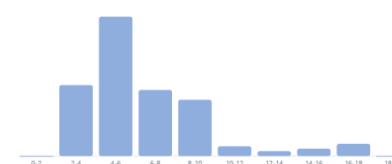
Placement Rate by Branch



Students Placed by Company (Top 8)



Package Distribution (LPA)



02 Skill Demand vs. Readiness

What does industry need, and are we ready?

02 **Skill Demand vs. Readiness**

What does industry need, and are we ready?

SKILL **Python** SQL Power BI Machine Learning Java Communication [Reset](#)

CORE ANALYTIC

Skill Gap = Industry Demand – Student Readiness

SQL 86.0% demand – 49.8% readiness = 36.2 percentage points

A positive gap indicates a readiness shortfall relative to the demonstration demand level.

HIGHEST PRIORITY SKILL

SQL

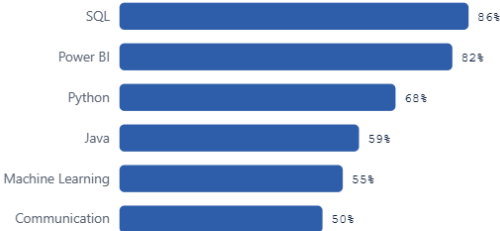
DEMAND
86.0%

READINESS
49.8%

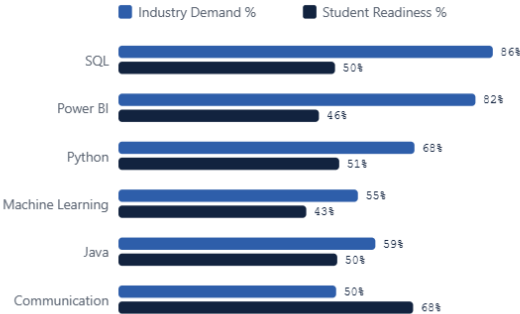
GAP
36.2 pts

CRITICAL

Industry Demand % by Skill



Demand % vs. Readiness % by Skill



WEB DASHBOARD — LIVE SITE

Action Center — Priority Matrix & Recommendation Engine

Placement Skill-Gap Hub

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Command Center

Skill Gap

Action Center

SAMPLE / DEMONSTRATION DATASET

03

Action Center

What should we do next? — uses the same Skill filter as above.

2-minute evaluator path

Focus on the largest gap, then show the recommended action.

Run demo path

01 Measure

Placement outcomes

→

02 Compare

Demand vs readiness

→

03 Prioritize

Largest positive gap

→

04 Act

Training recommendation

Skill Gap Priority Matrix

CRITICAL TRAINING PRIORITY

MAINTAIN

LOW PRIORITY

MONITOR

Industry Demand %

Student Readiness %

SQL

Power BI

Python

Java

Machine Learning

Communication

● Critical training priority

● High

● Medium / Monitor

● Low / Maintain

X: Student Readiness % · Y: Industry Demand % · High demand + low readiness (top-left) = highest priority.

Training Recommendation Engine (Rule-Based)

SKILL	PRIORITY	RECOMMENDATION
SQL	CRITICAL	Prioritize SQL training immediately — largest readiness gap.
Power BI	CRITICAL	Prioritize Power BI training immediately — largest readiness gap.
Python	HIGH	Schedule Python training this term — meaningful shortfall.
Machine Learning	MEDIUM	Reinforce Machine Learning through practice — modest shortfall.
Java	MEDIUM	Reinforce Java through practice — modest shortfall.
Communication	LOW	Maintain current Communication readiness — no urgent gap.

SWITCH statement on Skill Gap % thresholds — not AI or machine learning.

CODEAVENGERS 2K26 — Master Report

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III.03

GitHub Repository

Full source: strategy docs, dataset, generator scripts, Power BI file, and web dashboard

-
- `codeavengers-placement-skill-gap`

Repository — codeavengers-placement-skill-gap

☰

 jonekavish-dot / codeavengers-placement-skill-gap

🔍 Type / to search



<> Code

🔗 Issues

🔗 Pull requests

🔗 Agents

🔗 Actions

🔗 Projects

🔗 Wiki

🔗 Security and quality

🔗 Insights

⚙️ Settings

 codeavengers-placement-skill-gap Public

📌 Pin

👁 Watch 0

🍴 Fork 0

★ Star 0

main

1 Branch

0 Tags

🔍 Go to file

Add file

<> Code

 jonekavish-dot Updated DashBoard UI intro page, Updated README and PDFs ✓ b488a89 · 32 minutes ago 3 Commits

01_Strategy

Updated DashBoard UI intro page, Updated README and PDFs

32 minutes ago

02_Dataset

Initial release: Placement Skill-Gap Hub

1 hour ago

03_Dataset_Generator_Scripts

Initial release: Placement Skill-Gap Hub

1 hour ago

PowerBI

Updated PowerBI UI insertion

1 hour ago

.gitignore

Initial release: Placement Skill-Gap Hub

1 hour ago

README.md

Updated DashBoard UI intro page, Updated README and PDFs

32 minutes ago

index.html

Updated DashBoard UI intro page, Updated README and PDFs

32 minutes ago

README

Placement Skill-Gap Hub

From Placement Statistics to Skill-Gap Intelligence.

A data-driven placement analytics and decision-support solution built for HACKORBIT 2K26 – Track A: DataDrishTi by Team Codeavengers, BANNARI AMMAN INSTITUTE OF TECHNOLOGY.

About

Power BI-based placement analytics that compares industry skill demand with student readiness to identify skill gaps and training priorities. Built by Team CODEAVENGERS for HACKORBIT 2K26.

[codeavengers-placement-skill-gap.ve...](#)

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★ 0 stars

👁 0 watching

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Deployments 3

✓ Production 32 minutes ago

Packages

No packages published
[Publish your first package](#)

III.04

Vercel Deployment

The web dashboard, live and publicly reachable — no install required to view

-
- codeavengers-placement-skill-gap.vercel.app

VERCEL

Production Deployment

jonekavish-dot...Hobby

Find

Overview

Deployments

Logs

Analytics

Speed Insights

Observability

Firewall

CDN

Environment Variables

Domains

ConnectBeta

Integrations

Storage

Flags

Agent

AI Gateway

Sandboxes

Workflows

ImagesBeta

jonekavish-dot

codeavengers-placem...Deployments / HTt1nz1mW

DeploymentLogsResourcesSourceOpen Graph

Deployment Details

Placement Skill-Gap Hub

Created

jonekavish-...30m ago

Status

Ready Latest

Duration

1s30m ago

Environment

ProductionCurrent

Domains

codeavengers-placement-skill-gap.vercel.app

codeavengers-placement-skil-git-5fff07-jonekavish-dots-projects.vercel.app

codeavengers-placement-skill-izg255e88-jonekavish-dots-projects.vercel.app

Source

main

b488a89 Updated DashBoard UI intro page, Updated README and PDFs

Deployment Settings

3 Recommendations

Deploy Logs

Deployment Summary

Deployment Checks

Assigning Custom Domains

Runtime Logs

View and debug runtime logs & errors

Observability

Monitor app health & performance

Speed Insights

Not Enabled

Performance metrics from real users

Web Analytics

Not Enabled

Analyze visitors & traffic in real-time